

# Milestone 1

## Objective: Movie Revenue Prediction

The goal of this project is to develop an **Interactive dashboard** based on machine learning model that accurately predicts a movie's box office revenue based on various features such as budget, genre, director, cast, critic/audience ratings, and historical box office performance.

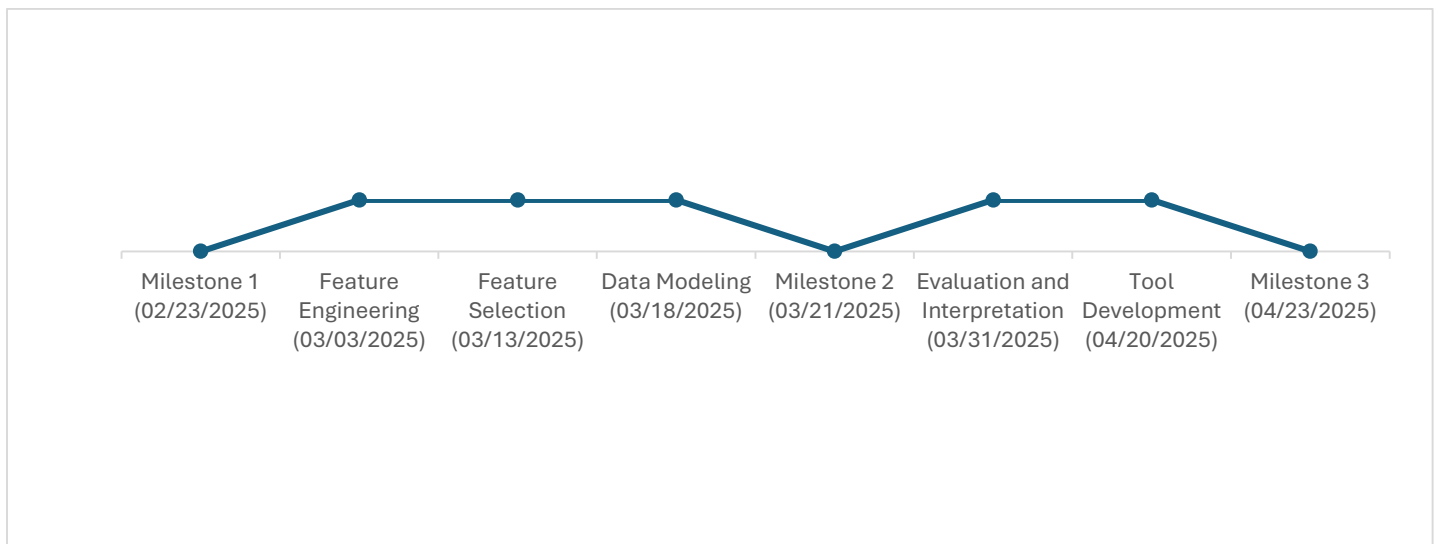
Our aim is to identify major factors that influence a film's success at the Box Office, by leveraging data from sources like IMDb, TMDb, and Rotten Tomatoes.

## Tech Stack:

- **Python:** Main programming language for data processing & modeling
- **Pandas, NumPy, SciPy, sklearn:** Data manipulation and preprocessing
- **SQLite:** Store cleaned and structured data
- **Matplotlib, Seaborn, Plotly:** Data visualization and EDA
- **Scikit-Learn, Pytorch:** Machine Learning
- **Flask:** API for serving predictions
- **Streamlit:** Interactive web-based dashboard for model insights

**Audience:** filmmakers, producers, and investors.

## Project Timeline:



## EDA report:

This Exploratory Data Analysis (EDA) report focuses on the dataset sourced from the Kaggle platform (TMDb and Rotten Tomatoes), IMDb.

These dataset sizes vary from 20000 to a million records of movie metadata like budget, revenue, audience ratings, critic ratings, movie crew, genre, popularity etc. This data is vital for understanding revenue trends, audience interest, and budget involvement.

The primary goal of this EDA is to uncover underlying patterns, identify potential anomalies or outliers, and gather insights that could help optimize sales strategies and improve overall business performance. Specifically, the objectives are:

1. To provide a comprehensive summary of the central tendencies and distributions of key variables.
2. To explore relationships between revenue and other factors such as budget, audience interest, genre, region, and critics rating and few others.
3. To identify trends over time.
4. To detect anomalies that could signify data issues or exceptional events.

## Dataset Description:

| Explored By     | Dataset                | Source | Variables                                                                                                                                                                                                                                                                                                                                                                                                       | Dimension                |
|-----------------|------------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| Mohammad Wael   | TMDb                   | Kaggle | id, title, vote_average, vote_count, status, release_date, revenue, runtime, adult, backdrop_path, budget, homepage, imdb_id, original_language, original_title, overview, popularity, poster_path, tagline, genres, production_companies, production_countries, spoken_languages, keywords                                                                                                                     | 1181912 Rows, 24 Columns |
|                 | Rotten Tomatoes Critic | Kaggle | rotten_tomatoes_link, critic_name, top_critic, publisher_name, review_type, review_score, review_date, review_content                                                                                                                                                                                                                                                                                           | 1130017 Rows, 8 Columns  |
|                 | Rotten Tomatoes Movie  |        | rotten_tomatoes_link, movie_title, movie_info, critics_consensus, content_rating, genres, directors, authors, actors, original_release_date, streaming_release_date, runtime, production_company, tomatometer_status, tomatometer_rating, tomatometer_count, audience_status, audience_rating, audience_count, tomatometer_top_critics_count, tomatometer_fresh_critics_count, tomatometer_rotten_critics_count | 17712 Rows, 22 Columns   |
|                 | Title.Basics           | IMDb   | tconst, titleType, primaryTitle, originalTitle, isAdult, startYear, endYear, runtimeMinutes, genres,                                                                                                                                                                                                                                                                                                            | 11464895 Rows, 9 Columns |
| Vishal K. Setti | Name.Basics            |        | nconst, primaryName, birthYear, deathYear, primaryProfession, knownForTitles                                                                                                                                                                                                                                                                                                                                    | 14195120 Rows, 6 Columns |
|                 | Title.Rating           |        | tconst, averageRating, numVotes                                                                                                                                                                                                                                                                                                                                                                                 | 1535911 Rows, 3 Columns  |
|                 | Title.Principals       |        | tconst, ordering, nconst, category, job, characters                                                                                                                                                                                                                                                                                                                                                             | 90984102 Rows, 6 Columns |
|                 | Title.Crew             |        | tconst, directors, writers                                                                                                                                                                                                                                                                                                                                                                                      | 11464895 rows, 3 Columns |
|                 | ISO2 codes to ISO3     | Kaggle | country, alpha2, alpha3, numeric                                                                                                                                                                                                                                                                                                                                                                                | 249 Rows, 4 Columns      |

## Pre-Processing:

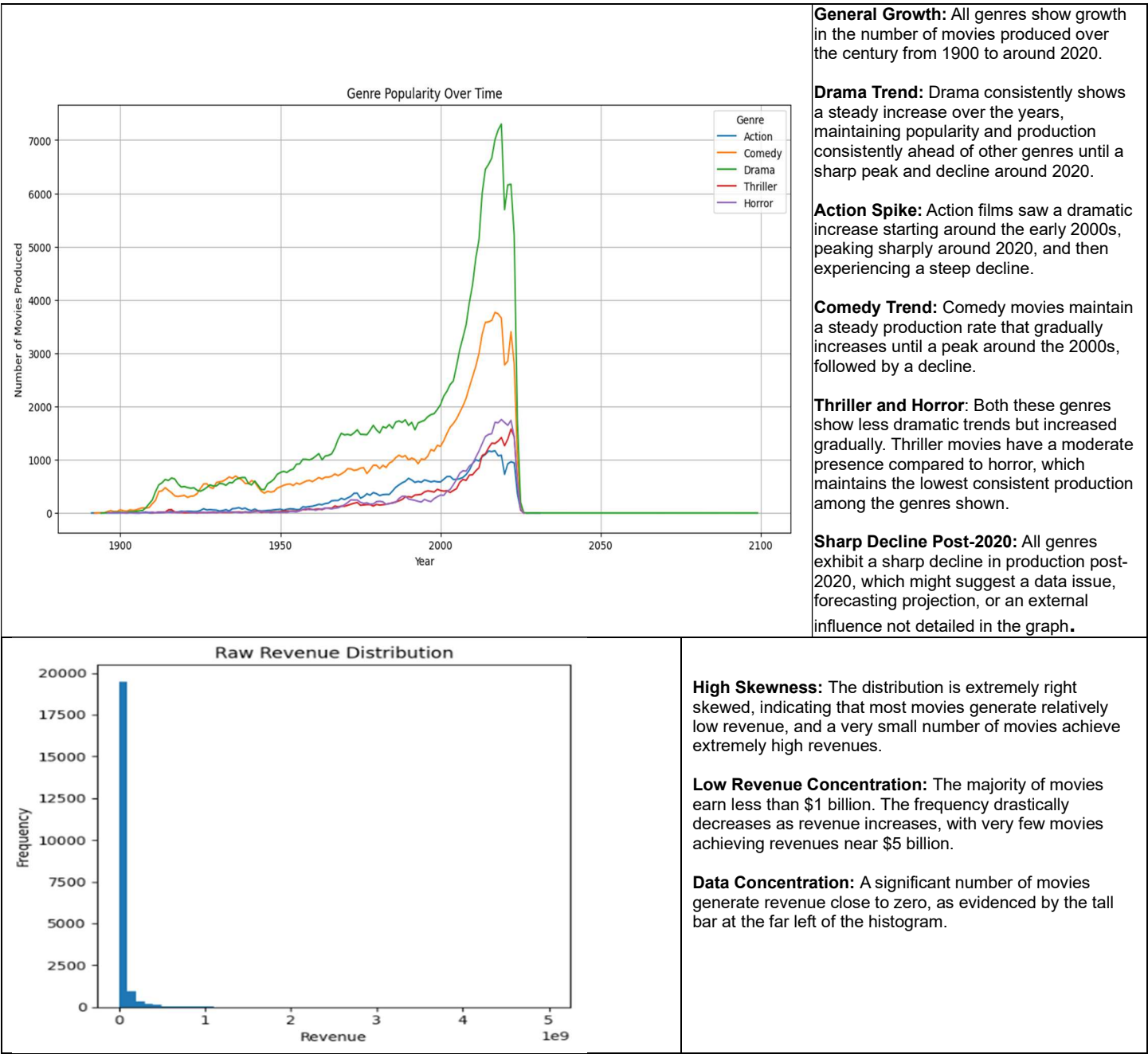
This exploration has incorporated various data cleaning methods such as imputation (arithmetic mean for numeric types and mode for categorical types) if the available data is more than 70%, otherwise dropping rows, and slicing columns as per the requirement. Irrelevant columns that are not being used were dropped.

Outliers were handled, using IQ, Z Square, and MAD methods, where the outliers were highlighted by any of the two methods, added a cap on the field. For skewed fields, log transformation and log transformation followed by cap were used.

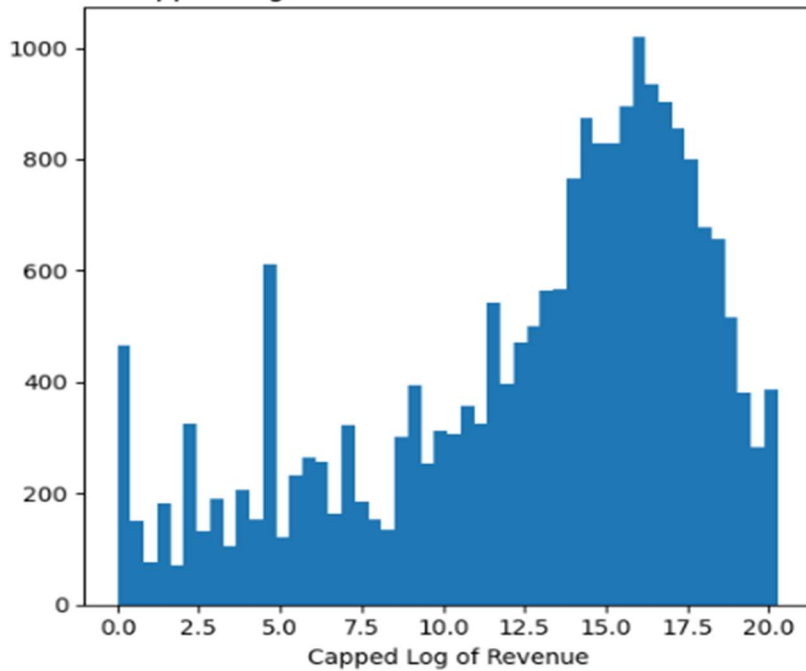
Few data transformations have been performed, in cases where the single column contains mixed type ex: review score contains data in format 1/5, 1/10/ 2.5/ 4, A+, D and so on, such data was transformed into single numeric type data.

In some cases, a mapping table was required to map distribution of revenue over the regions where the movie was released. To map the fields from different datasets, the datasets were merged on key parameters.

Univariate Analysis:



### Capped Log-Transformed Revenue Distribution



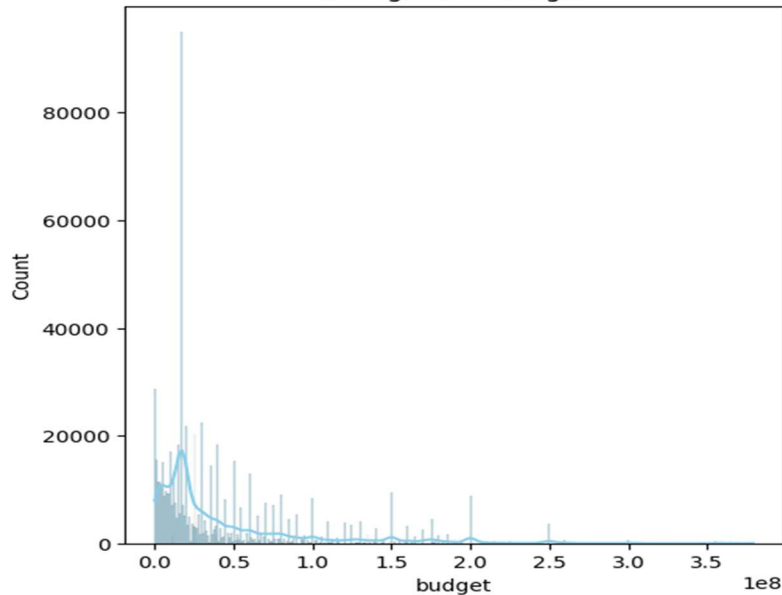
**Bell Shape:** Unlike the previous raw revenue distribution, which was highly skewed, this histogram exhibits a more bell-shaped curve. This indicates that the log transformation has normalized the distribution, making it easier to analyze statistically.

**Mode Concentration:** The highest frequency of movies appears to be clustered around the middle of the revenue range (around 10 on the log scale), suggesting that this is the most common revenue scale when logarithmic adjustments are applied.

**Symmetry:** Post-transformation, the distribution appears relatively symmetric with tails extending towards both lower and higher revenue scales.

**Diverse Revenue Ranges:** The histogram shows that movies' revenues span a wide range but are more consistently distributed in logarithmic terms. Smaller peaks present at lower capped log value suggest there are still significant numbers of lower-revenue films.

### Histogram of budget



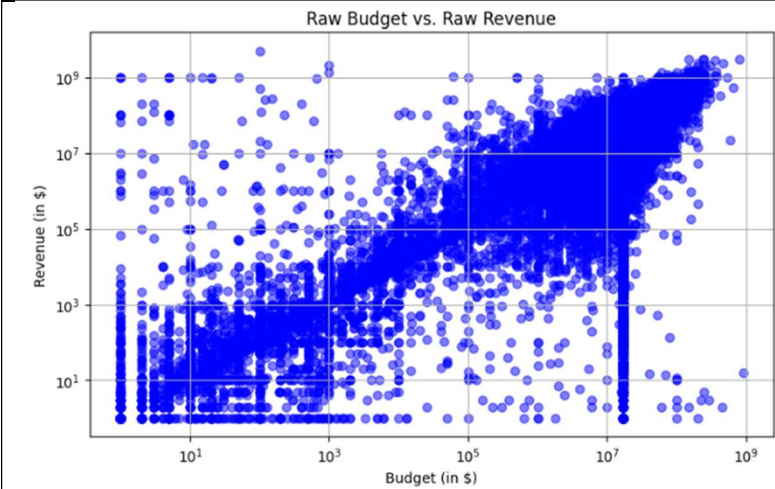
**High Skew to the Right:** The distribution is right-skewed, indicating that while most movies have relatively low budgets, a few have very high budgets extending to the far right of the histogram.

**Peak at Lower Budgets:** There is a prominent peak at the lower end of the budget scale, showing that a significant majority of movies operate on modest budgets. The count rapidly decreases as the budget increases.

**Tail Distribution:** The long tail extending towards higher budget values suggests that although rare, some movies are produced with substantial financial investment, over \$150 million.

**Budget Dynamics:** The concentration of data points near the lower end of the budget scale and the sparse distribution towards the higher end reflect the economic realities of the film industry, where high budget films are less common due to their risky financial nature.

Bivariate or Multivariate Analysis:



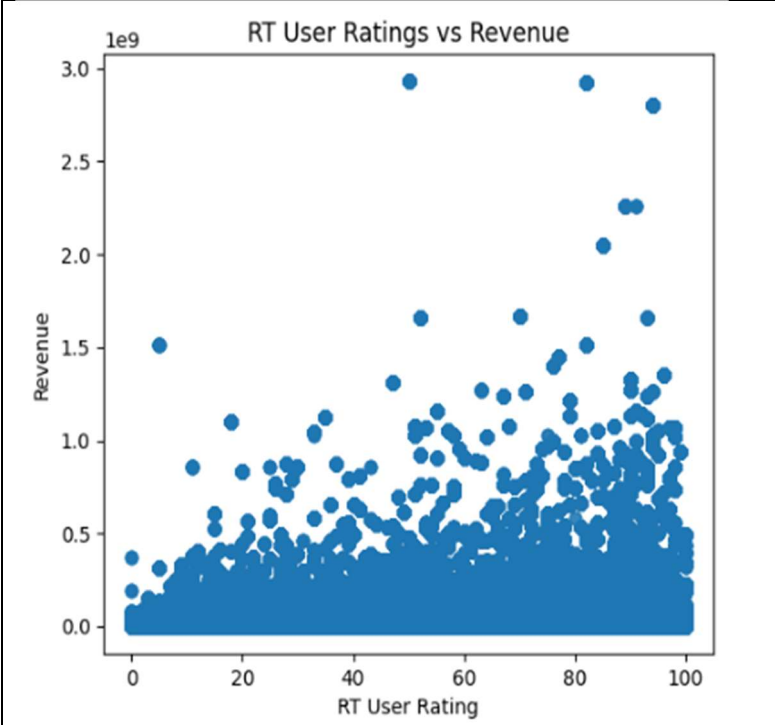
**Logarithmic Scale:** Both budget and revenue are displayed on logarithmic scales to manage the wide range of values and enhance visibility across different magnitudes.

**Positive Correlation:** There is a general positive correlation, indicating that higher budgets usually lead to higher revenues, especially noticeable in films with budgets over \$100,000.

**Variable Outcomes for Low Budgets:** While the density of data points is high for lower, the resulting revenues vary significantly, suggesting other factors besides budget influence revenue outcomes.

**Outliers:** Some films with moderate budgets achieve exceptionally high revenues, and others with large budgets do not perform as well, highlighting that a bigger budget doesn't always guarantee higher revenue.

**High Budget Efficiency:** Films with budgets above \$10 million often show a clear trend where revenues significantly exceed the investment, indicating the potential success of blockbuster investments.



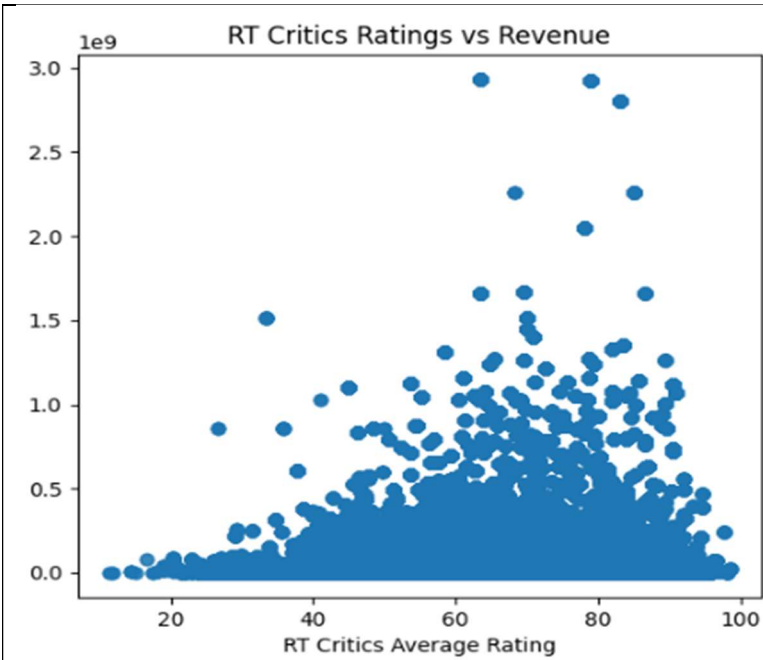
**Weak Correlation:** The scatter plot does not show a strong or clear linear correlation between user ratings and revenue. The data points are widely dispersed around the plot, which suggests that higher user ratings do not consistently predict higher revenues.

**Revenue Distribution:** Most of the movies, irrespective of the RT user rating, tend to generate revenue of 1 billion dollars or less. There are only a few outliers that surpass the 1 billion mark, indicating that very high revenues are rare and not strongly dependent on user ratings.

**Cluster of Values:** A large cluster of movies rated between 40 and 80, with revenues below 0.5 billion, hints at a general tendency but lacks a definite pattern indicating that ratings within this broad range do not significantly impact revenue.

**Extreme Values:** Movies with very high revenues are scattered across a wide range of user ratings, from moderately low to very high (20 to 100). This shows that some movies can be highly successful in terms of revenue even without exceptionally high Rotten Tomatoes user ratings.

**High Ratings and Revenues:** There are several outliers with both high ratings (above 80) and high revenues (above 1 billion), suggesting that while not a firm rule, higher user satisfaction might contribute to higher earnings for certain films.



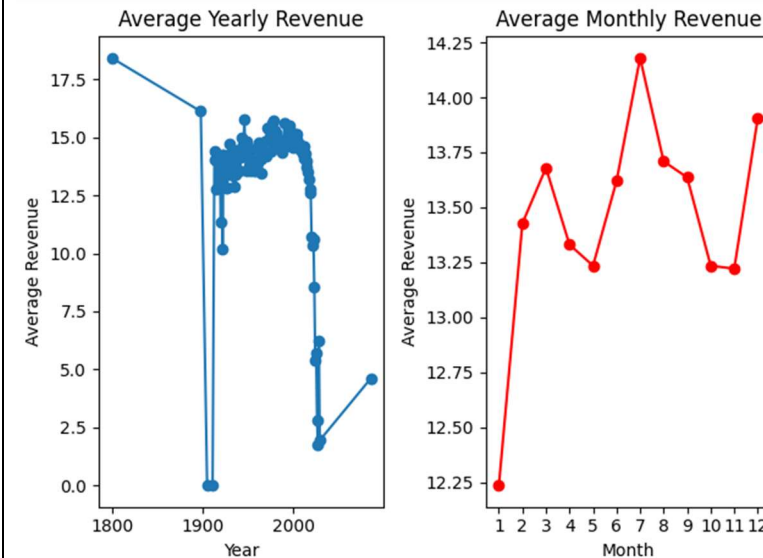
**Distribution of Data Points:** The scatter plot illustrates that most movies, regardless of the critics' ratings, generate revenues under 1 billion dollars. The dispersion of data across critic ratings suggests there is no strong linear correlation between high ratings and high revenue.

**High Revenue Outliers:** A few movies achieve revenues exceeding 1.5 billion dollars, and these are spread across a wide range of critics' ratings from 40 to 100. This indicates that while critical acclaim may be advantageous, it is not always a predictor of financial success.

**Clustering of Movies:** A significant cluster of movies with critics' ratings in the range of 40 to 80 and revenues below 0.5 billion dominates the plot. This dense concentration shows that most movies fall within this moderate rating and lower revenue bracket.

**Ratings Above 80:** Movies rated above 80 by critics show relatively better revenue performance compared to lower-rated movies, though the scatter remains broad. This suggests some degree of positive impact from higher critics' ratings on revenues, but with considerable variability.

**Low Rating, High Revenue:** There are movies with rather low critics' ratings (20-60) that have surprisingly high revenues. Such instances highlight that other factors like genre, audience appeal, star power, and marketing could heavily influence box office performance.

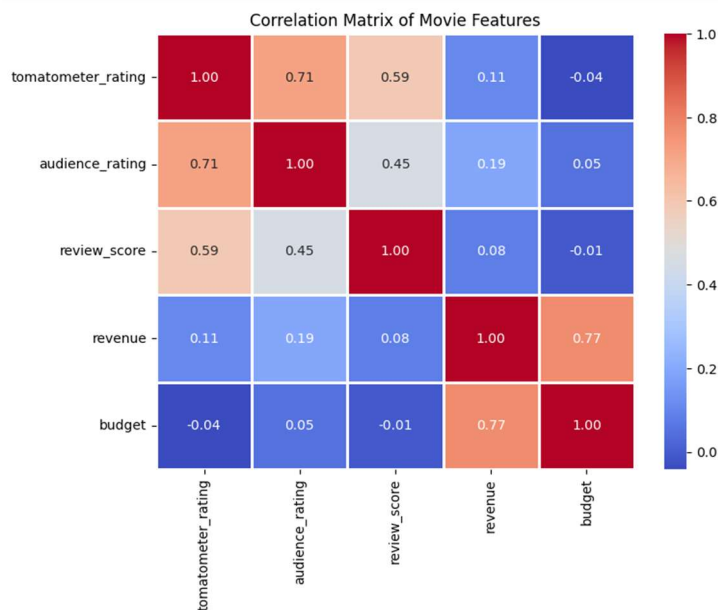


#### Average Yearly Revenue (Blue Line)

- Steady Highs Pre-1950:** The graph shows that the average yearly revenue was relatively high and stable before 1950.
- Significant Decline and Fluctuations:** From around 1950, there is a sharp decline followed by substantial fluctuations in revenue up to around 2000.
- Further Drop in the 2000s:** After 2000, revenue dipped lower, suggesting a possible disruption or change in the industry.
- Slight Recovery in Recent Years:** There is a slight upward trend in the last data points, hinting at some recovery or adaptation within the industry.

#### Average Monthly Revenue (Red Line)

- Notable Peaks:** The graph shows revenue peaks in March (3), July (7), and November (11), indicating higher earnings during these months which may correlate with school holidays and major holiday seasons.
- Dips in Revenue:** There are notable drops in months like May (5) and September (9), possibly linked to lesser major holidays or events during these periods.
- General Variability:** The plot shows variability from month to month, which can be attributed to consumer habits, movie release schedules, and seasonal industry trends.



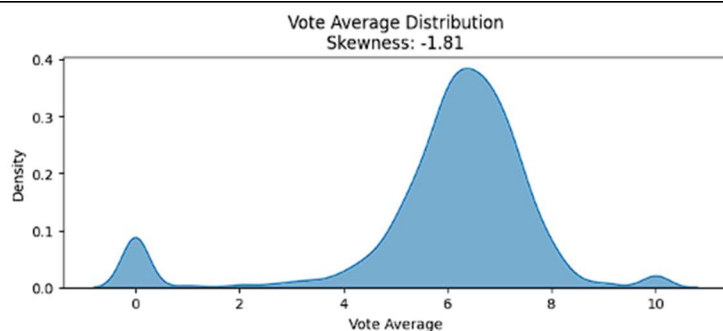
**Tomatometer Rating and Audience Rating:** There's a strong positive correlation (0.71) between tomatometer rating and audience rating, suggesting that movies which are well-received by critics are also likely to be favored by the general audience.

**Tomatometer Rating and Review Score:** A moderate positive correlation (0.59) is observed between tomatometer rating and review score, indicating that higher critic ratings often coincide with higher review scores.

**Audience Rating and Review Score:** There is a moderate positive correlation (0.45) between audience rating and review score, suggesting that audience perceptions are somewhat aligned with review scores.

**Revenue and Budget:** There is a strong positive correlation (0.77) between revenue and budget. This implies that movies with larger budgets tend to generate higher revenues, which could be indicative of higher production values attracting larger audiences.

**Revenue and Ratings:** The correlations between revenue and both tomatometer rating (0.11) and audience rating (0.19) are relatively low. This suggests that while there is some positive relationship between these ratings and revenue, many other factors are likely to influence a movie's financial success.

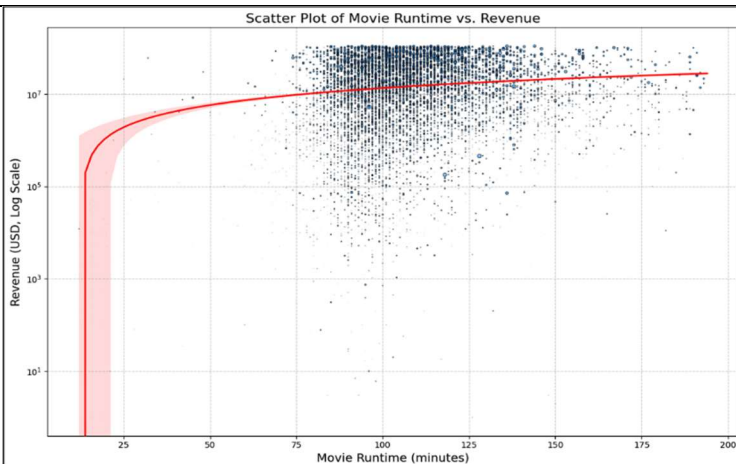


#### Skewness of Vote Average Distribution plot:

This graph shows a vote average distribution with a skewness value of -1.81. Main features are:

- A large peak centered around 6-7 on the vote average scale
- A smaller peak near 0
- A very small peak around 9-10
- The x-axis ranges from 0 to 10, representing vote averages
- The y-axis shows density, ranging from 0 to approximately 0.4

The negative skewness value (-1.81) is reflected in the graph's shape, showing that the distribution is skewed to the left, with the bulk of the data concentrated on the right side and a longer tail extending to the left. The concentration of votes around 6-7 suggests that most items receive above-average ratings.



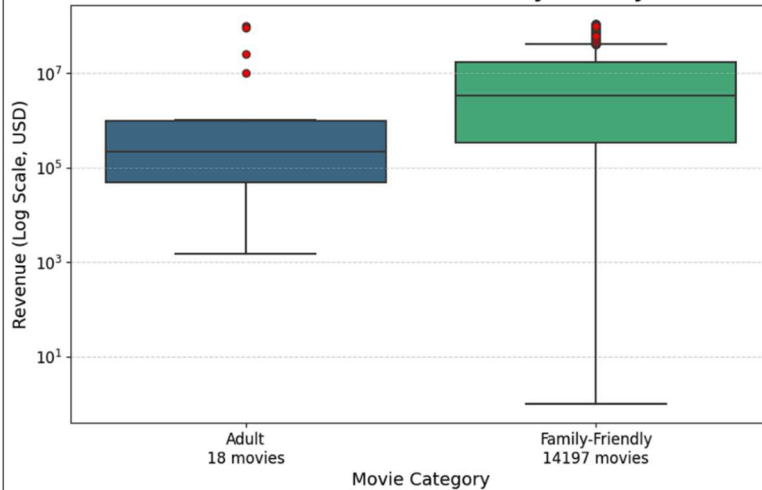
#### Movie runtime Scatter Plot:

The scatter plot visualizes the relationship between movie runtime (x-axis) and revenue (y-axis, log scale):

- A positive correlation exists, where longer movies tend to have higher revenue, but the effect diminishes after a certain runtime.
- The red regression curve with a shaded confidence interval suggests a logarithmic trend, meaning revenue increases rapidly for shorter movies but stabilizes beyond ~100 minutes.
- There is high variance in revenue for all runtimes, indicating other influencing factors.
- Outliers exist, with some shorter movies having high revenue and longer movies earning less.

This analysis suggests that while longer runtimes may contribute to higher revenue, other factors likely play a significant role.

**Revenue Distribution: Adult vs. Family-Friendly Movies**



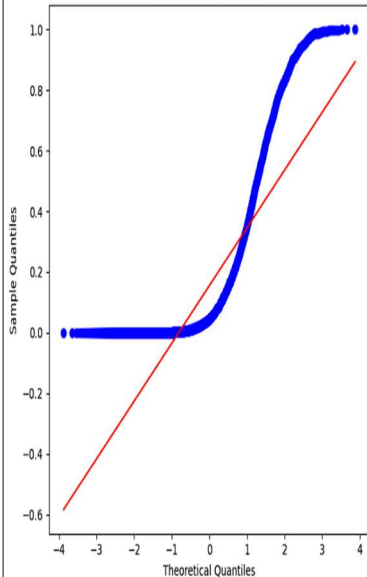
**Revenue distribution of Adult vs Family-Friendly movies:**

This box plot compares revenue distribution between Adult and Family-Friendly movies, shown on a logarithmic scale (USD):

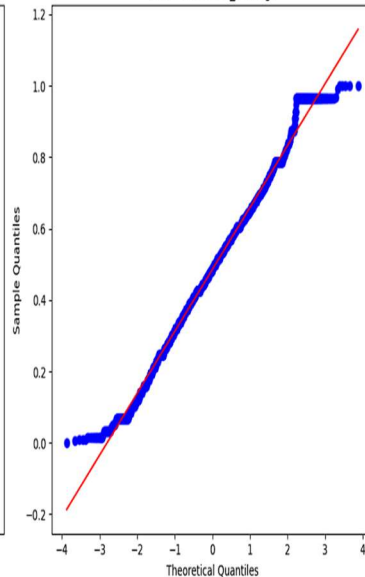
- Family-Friendly movies (14,197 films) show a significantly larger sample size compared to adult movies (18 films)
- Family-Friendly movies have a higher median revenue and wider revenue range
- Adult movies show a more concentrated revenue distribution with some outliers at the top
- Both categories display some outliers, indicated by red dots
- The y-axis uses a logarithmic scale, ranging from  $10^1$  to  $10^7$  USD

The graph clearly illustrates that Family-Friendly movies dominate both in terms of quantity and potential revenue generation

Q-Q Plot of revenue



Q-Q Plot of vote\_average



**Skewness of movie director influence:**

The image shows two Q-Q (Quantile-Quantile) plots side by side:

Left Plot - Q-Q Plot of Revenue:

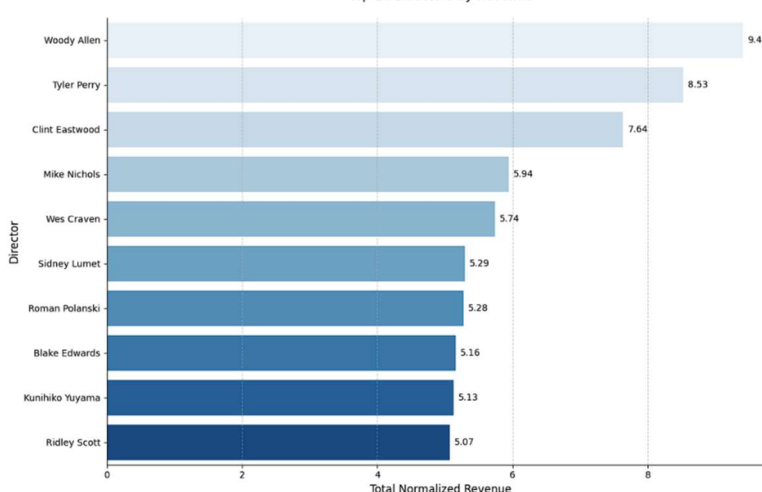
- Data deviates significantly from the red diagonal line (theoretical normal distribution)
- Indicates the revenue distribution is not normal and heavily skewed
- Has a long flat portion at the bottom and steep rise in the middle
- Shows clustering at both extremes

Right Plot - Q-Q Plot of Vote Average:

- Shows a more linear pattern following the red reference line
- Data points (in blue) closely follow the theoretical normal distribution line
- Some minor deviations at the extremes
- Range approximately between 0 and 1 on the sample quantiles axis
- Suggests vote averages are more normally distributed than revenue

The vote averages appear to be more normally distributed than the revenue data.

**Top 10 Directors by Revenue**

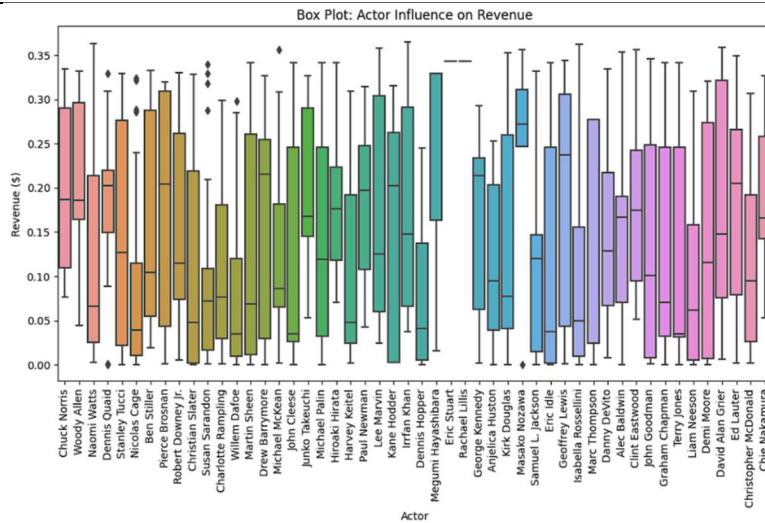


**Top 10 Directors by Revenue:**

The bar graph shows total normalized revenue on the x-axis and director names on the y-axis. The directors are ranked as follows:

- Woody Allen leads with the highest normalized revenue of 9.41
- Tyler Perry follows with 8.53
- Clint Eastwood ranks third with 7.64
- Mike Nichols has 5.94
- Wes Craven shows 5.74
- Sidney Lumet earned 5.29
- Roman Polanski with 5.28
- Blake Edwards achieved 5.16
- Kuniyiko Yuyama at 5.13
- Ridley Scott with 5.07

The graph uses a gradient of blue shades, transitioning from lighter blues for the top-ranking directors to darker blues for those lower on the list. The bars are horizontally oriented, making it easy to compare the revenue values between directors. Grid lines are included to help with precise value readings.



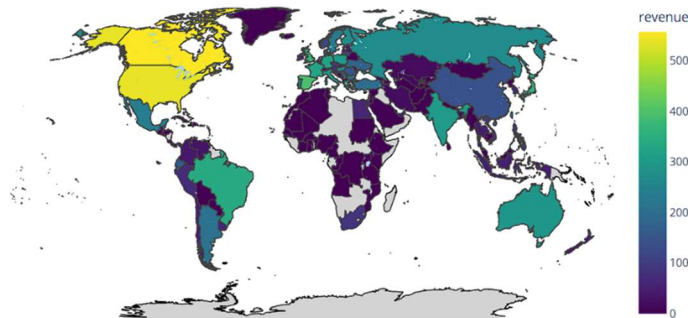
### Actors influence on revenue:

This box plot visualizes the distribution of revenue influence across different actors in the entertainment industry. The y-axis shows revenue values ranging from 0 to approximately 0.35, while the x-axis lists various actor names.

- Most actors show median revenue influences between 0.10 and 0.25
- The highest median values appear to be around 0.30-0.35
- There is considerable variation in revenue influence for most actors, as shown by the box sizes and whiskers
- Several actors show outliers, represented by diamond-shaped points above their boxes
- The whiskers extend from approximately 0 to 0.35, showing the full range of revenue influence
- The interquartile ranges (boxes) show the middle 50% of revenue values
- Outliers generally appear above the upper whiskers, indicating some exceptional high-revenue performances

The plot provides a comprehensive view of how different actors impact revenue, with some showing higher median values but also greater variability.

Total Revenue by Region



### Total revenue by region (choropleth plot):

#### Revenue Distribution

- Highest revenue (≈500): North America (USA and Canada) shown in yellow
- Moderate-high revenue (≈300-400): Western Europe and Australia shown in teal/turquoise
- Medium revenue (≈200): Parts of Asia shown in blue
- Low revenue (≈0-100): Most of Africa and Middle East shown in dark purple

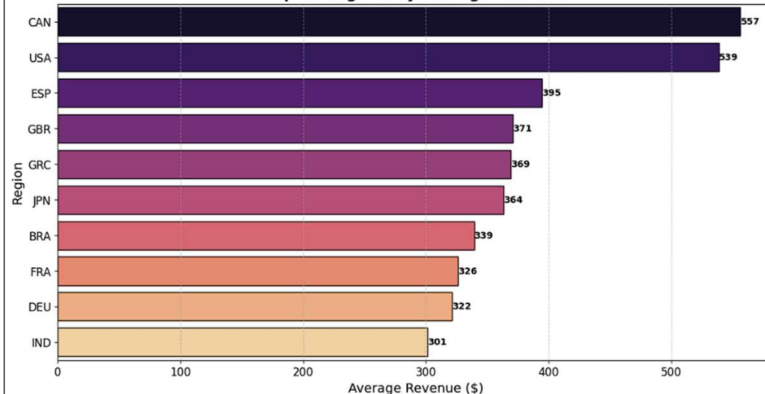
#### Regional Highlights

- Brazil and parts of South America show moderate revenue levels
- Russia displays medium-low revenue values
- India shows moderate revenue levels

#### Scale Details

- Revenue scale ranges from 0 to 500
- Color gradient goes from dark purple (lowest) to yellow (highest)
- Intermediate values shown in blues and teals

Top 10 Regions by Average Revenue



### Top 10 regions by Average Revenue:

#### Revenue Leadership:

- North American dominance with Canada (\$557) and USA (\$539) having significantly higher revenues
- These two countries are the only ones generating above \$500 in average revenue

#### Regional Distribution:

- Europe has strong representation with 5 countries: Spain, Great Britain, Greece, France, and Germany
- Asia is represented by Japan and India
- South America has single representation through Brazil
- No African or Australian regions in top 10

#### Economic Implications:

- Developed economies generally show higher average revenue
- Emerging market India ranks last despite its large market size
- European market shows consistent performance in middle-range revenue generation

## Insights and Findings:

1. **Industry Evolution Impact:** The significant fluctuations and general decline in yearly revenues post-1950 highlight the impact of technological changes and new media platforms on traditional revenue streams. Recent years show a slight recovery, suggesting effective adaptation strategies.
2. **Strategic Seasonality:** Monthly revenue data reveal clear peaks during months such as March, July, and November, aligning with school vacations and major holidays, emphasizing the importance of strategic release timing to capitalize on these high-revenue periods.
3. **Adaptation and Strategic Planning:** Despite historical challenges, the entertainment industry demonstrates resilience, and the data suggests that strategic planning regarding release scheduling could optimize revenue outcomes in response to established seasonal consumer behavior patterns.
4. **Market Dominance:** Family-friendly movies vastly outnumber adult films (14,197 vs 18) and show higher revenue potential, with both categories displaying notable outliers in the high-revenue range.
5. **Director Performance:** Woody Allen leads revenue generation (9.41), followed by Tyler Perry (8.53) and Clint Eastwood (7.64), demonstrating that established directors with distinct styles tend to generate higher revenues.
6. **Global Distribution & Runtime:** North America shows highest revenue concentration, with Europe and Australia following; movie revenues generally increase with runtime up to 150 minutes, after which returns plateau, suggesting an optimal length window for maximum revenue potential.

## References:

### Data sets:

1. **IMDb**
  - a. **Provider:** IMDb
  - b. **Dataset URL:** <https://developer.imdb.com/non-commercial-datasets/>
2. **TMDb:**
  - a. **Contributed by:** asaniczka
  - b. **Profile URL:** <https://www.kaggle.com/asaniczka>
  - c. **Dataset URL:** <https://www.kaggle.com/datasets/asaniczka/tmdb-movies-dataset-2023-930k-movies>
3. **Rotten Tomatoes**
  - a. **Contributed by:** Stefano Leone
  - b. **Profile URL:** <https://www.kaggle.com/stefanoleone992>
  - c. **Dataset URL:** <https://www.kaggle.com/code/stefanoleone992/rotten-tomatoes-eda>
4. **Country Codes ALPHA-2 & ALPHA-3**
  - a. **Contributed by:** emolodov
  - b. **Profile URL:** <https://www.kaggle.com/emolodov>
  - c. **Dataset URL:** <https://www.kaggle.com/datasets/emolodov/country-codes-alpha2-alpha3/data>

## LLM Utilization:

During the development of our **Movie Revenue Prediction Model**, we sought clarification and guidance from an LLM on various aspects of data analysis, feature selection, and visualization. Below are the prompts we used and how the responses influenced our work:

### 1. Data Cleaning and Preprocessing

- **Prompt:** "I have datasets with missing values in budget, runtime, and revenue. What are some effective strategies to handle these missing values while maintaining data integrity?"
- **Utilization:** We used median imputation for budget and runtime while only considering movies with revenue > 0 instead of imputing revenue values, based on the suggestions.

### 2. Outlier Detection and Normalization

- **Prompt:** "My dataset contains extreme values in budget and revenue. What are common statistical methods to detect and handle such outliers?"
- **Utilization:** We applied **Z-scores** and **IQR** to detect and filter extreme values, ensuring the model was not skewed by anomalies.

### 3. Visualizing Trends and Relationships

- **Prompt:** *"Want to analyze the correlation between movie runtime and revenue using scatter and line plots, how to approach this?"*
- **Utilization:** This helped us implement **scatter and line plots** to observe revenue trends concerning runtime, leading to key insights about optimal movie lengths.

### 4. Feature Selection for Revenue Prediction

- **Prompt:** *"I have datasets containing details like cast, directors, language, and budget, how do I approach the movie revenue analysis with these? "*
- **Utilization:** The response helped us prioritize budget, director influence, actor influence, and language/region impact as key features for revenue analysis.